chunyuac_HW4

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1.

(a)

The likelihood function is

$$L(\alpha) = \prod_{i=1}^{n} \frac{\alpha \lambda^{\alpha}}{X_i^{\alpha+1}},$$

so the log likelihood is

$$l(\alpha) = \sum_{i=1}^{n} (\log \alpha + \alpha \log \lambda - (\alpha + 1) \log X_i)$$

and its derivative with respective to α

$$\begin{aligned} \frac{dl}{d\alpha} &= \sum_{i=1}^{n} \left(\frac{1}{\alpha} + \log \lambda - \log X_i \right) \\ &= \frac{n}{\alpha} + n \log \lambda - \sum_{i=1}^{n} \log X_i. \end{aligned}$$

Setting to zero and solve for α, we obtain the MLE

$$\hat{a} = \frac{1}{\frac{1}{n} \sum_{i=1}^{n} \log X_i - \log \lambda}.$$

(b)

The Fisher Information is

$$\begin{aligned}
I(\alpha) &= E\left(-\frac{\partial^2 \log f(X;\alpha)}{\partial\alpha^2}\right) \\
&= E\left(-\frac{\partial^2}{\partial\alpha^2}(\log\alpha + \alpha\log\lambda - (\alpha+1)\log X)\right) \\
&= E\left(-\frac{\partial}{\partial\alpha}\left(\frac{1}{\alpha} + \log\lambda - \log X\right)\right) \\
&= E\left(\frac{1}{\alpha^2}\right) = \frac{1}{\alpha^2},
\end{aligned}$$

so for large n, we have the approximation

$$\hat{\alpha} \approx N\left(\alpha, \frac{1}{nI(\alpha)}\right) = N(\alpha, \alpha^2/n).$$

In practice when α is unknown, we can replace a by an estimate given by the MLE $\hat{a}$.

(c)

The distribution of α is approximately $N(\hat{\alpha}, \hat{\alpha}^2/n)$. Now given $\lambda = 2$ and a sample of size $n = 100$ such that $\sum_i \log(x_i) = 107.39$, we have an estimate

$$\hat{\alpha} = \frac{1}{\frac{1}{n}\sum_{i=1}^n \log x_i - \log\lambda} = \frac{1}{\frac{107.39}{100} - \log 2} = 2.626.$$

[1]:
```
from math import log

1/(1.0739 - log(2))
```

[1]: 2.626375824269999

Thus the distribution of the MLE $\hat{\alpha}$ can be approximated by

$$N(\hat{\alpha}, \hat{\alpha}^2/n) = N(2.626, 0.2626^2).$$

The MLE $\hat{\alpha}$ is the best estimator for α in the sense that it is asymptotically efficient, meaning its bias vanishes as n goes to infinity (i.e., asymptotic unbiasedness), and among all asymptotically unbiased estimators, it achieves the lowest possible variance. Given the sample data and the above approximation, our best estimate for α is 2.626 and the standard error of the estimator $\hat{\alpha}$ is approximately 0.2626.

2.

(a)

The pdf is

$$f(x;\theta) = \frac{\beta^\theta}{\Gamma(\theta)} x^{\theta-1} e^{-\beta x}, \quad 0 < x < 1, \theta, \beta > 0,$$

so the Fisher Information is

$$\begin{aligned}
I(\theta) &= E\left(-\frac{\partial^2 \log f(X;\theta)}{\partial\theta^2}\right) \\
&= E\left(-\frac{\partial^2}{\partial\theta^2}(\theta \log\beta + (\theta-1)\log X - \log\Gamma(\theta) - \beta X)\right) \\
&= E\left(\frac{\partial}{\partial\theta}\frac{\Gamma'(\theta)}{\Gamma(\theta)}\right) \\
&= \frac{\Gamma(\theta)\Gamma''(\theta) - (\Gamma'(\theta))^2}{\Gamma^2(\theta)}.
\end{aligned}$$

(b)

Let θ_0 be the true value of θ and $\hat{\theta}$ its MLE. When n is large, we have approximately

$$\begin{aligned}
\hat{\theta} &\approx N\left(\theta_0, \frac{1}{nI(\theta_0)}\right) \\
&= N\left(\theta_0, \frac{\Gamma^2(\theta_0)}{n(\Gamma(\theta_0)\Gamma''(\theta_0) - (\Gamma'(\theta_0))^2)}\right).
\end{aligned}$$

Since θ_0 is unknown, so is this distribution. It can be thought of as a distribution family. Given a sample data, in order to obtain a distribution we can work with, the best we can do is to numerically find an estimate using the MLE and replace all θ_0 by this estimate.

(c)

By the invariance property, the MLE for $\log(\theta)$ is $\log(\hat{\theta})$, whose distribution can be approximated by a normal using the Delta Method. Let

$$\sigma_n = \sqrt{\frac{\Gamma^2(\theta_0)}{n(\Gamma(\theta_0)\Gamma''(\theta_0) - (\Gamma'(\theta_0))^2)}}.$$

Recall that $\Gamma^2(\theta_0)/(\Gamma(\theta_0)\Gamma''(\theta_0) - (\Gamma'(\theta_0))^2)$ is the second derivative of $\log\Gamma(\theta)$ at θ_0, which is finite given $\theta_0 > 0$. Thus we know that $\sigma_n \downarrow 0$ as $n \to \infty$. Since we have

$$\frac{\hat{\theta} - \theta_0}{\sigma_n} \xrightarrow{D} Z,$$

where $Z \sim N(0,1)$, by the Delta Method with $g(x) = 1/x$, we know that $(g'(\theta_0))^2 = 1/\theta_0^2$ and

$$\log(\hat{\theta}) \approx N\left(\log(\theta_0), \frac{\Gamma^2(\theta_0)}{n\theta_0^2(\Gamma(\theta_0)\Gamma''(\theta_0) - (\Gamma'(\theta_0))^2)}\right).$$

Again here θ_0 is unknown. Given a sample data we can replace it by a maximum likelihood estimate to obtain a workable distribution.

3.

(a)

By the invariance property, the MLE for the odds is

$$\frac{\hat{p}}{1-\hat{p}}.$$

(b)

Let $Y_1, Y_2, \ldots, Y_n$ be i.i.d. from the Bernoulli(p) distribution. We can write $X = \sum_{i=1}^n Y_i$ and $\hat{p} = \bar{Y}$. We also have

$$\begin{aligned}
E(\hat{p}) &= E(\bar{Y}) = E(Y_i) = p, \\
V(\hat{p}) &= V(\bar{Y}) = V\left(\sum_{i=1}^n Y_i/n\right) = V(Y_i)/n = p(1-p)/n.
\end{aligned}$$

By the CLT, we know that

$$\frac{\hat{p}-p}{\sqrt{\frac{p(1-p)}{n}}} \xrightarrow{D} Z.$$

{-} We want to apply the Delta Method with $g(x) = x/(1-x)$, which requires $g'(p) \neq 0$ and $\sigma_n = \sqrt{p(1-p)/n} \downarrow 0$ as $n \to \infty$. This is true for all $0 < p < 1$. We know that

$$g'(p) = \frac{(1-p)+p}{(1-p)^2} = \frac{1}{(1-p)^2}$$

and that by the Delta Method

$$\begin{aligned}
\frac{\hat{p}}{1-\hat{p}} &\approx N(g(p), (g'(p))^2\sigma_n^2) \\
&= N\left(\frac{p}{1-p}, \frac{1}{(1-p)^4}\frac{p(1-p)}{n}\right) \\
&= N\left(\frac{p}{1-p}, \frac{p}{n(1-p)^3}\right).
\end{aligned}$$

To obtain a workable distribution, plug in $p = \hat{p}$.

(c)

Given the above approximation (the one before plugging in $p = \hat{p}$), we have

$$P\left(\frac{\left|\frac{\hat{p}}{1-\hat{p}} - \frac{p}{1-p}\right|}{\sqrt{\frac{p}{n(1-p)^3}}} < Z_{\alpha/2}\right) = 1 - \alpha,$$

where $Z_{\alpha/2}$ is the $100(1 - \frac{\alpha}{2})$ percentile point of the standard normal distribution. Hence the confidence interval is

$$\frac{p}{1-p} \in \left[\frac{\hat{p}}{1-\hat{p}} - Z_{\alpha/2}\sqrt{\frac{p}{n(1-p)^3}}, \frac{\hat{p}}{1-\hat{p}} + Z_{\alpha/2}\sqrt{\frac{p}{n(1-p)^3}}\right]$$

Again this interval has an unknown p. To obtain a workable interval we plug in $p = \hat{p}$ and get

$$\left[\frac{\hat{p}}{1-\hat{p}} - Z_{\alpha/2}\sqrt{\frac{\hat{p}}{n(1-\hat{p})^3}}, \frac{\hat{p}}{1-\hat{p}} + Z_{\alpha/2}\sqrt{\frac{\hat{p}}{n(1-\hat{p})^3}}\right].$$

(d)

We look at the 95% confidence intervals. As the simulation results shown below, only around 65% of the 100,000 sample confidence intervals cover the true value of p, not 95%. The bar chart at the bottom shows the first 100 sample confidence intervals. The missing bars are intervals with length 0, which are simulation results with $\hat{p} = 0$. With $p = 0.1$ it is understandable that a sample value $\hat{p} = 0$ is not uncommon. Indeed, since the "rules of thumb" $np \geq 5, n(1-p) \geq 5$ are not satisfied, the CLT approximation applied in (b) is not expected to be accurate. So a simulation result like the below is to be expected.

[2]:
```
from scipy.stats import norm, binom
import numpy as np
from pandas import DataFrame
import matplotlib.pyplot as plt

n = 10
p = 0.1
alpha = 5
sampleSize = 100000
z = norm.ppf(1 - alpha/100/2)

phat = binom.rvs(n, p, size=sampleSize)/n
l = phat/(1-phat) - z*np.sqrt(phat/(n*(1-phat)**3))
r = phat/(1-phat) + z*np.sqrt(phat/(n*(1-phat)**3))
p = np.full(sampleSize, p)

((l < p) & (p < r)).sum()/sampleSize
```

[2]:
```
0.65566
```

[3]:
```
ax = DataFrame([l, r]).T.head(100).plot(kind='bar', stacked=True, legend=None,␣
↪color='y')
ax = DataFrame(p).head(100).plot(ax=ax, legend=None, figsize=(10, 4))
ax.set_xticks([])
ax.set_title("Simulation Results of Confidence Intervals and $p=0.1$")
plt.show()
```